ABC Phones Credit Portfolio Analysis

Nimble batch pipeline, enterprise-grade controls, and decision-ready portfolio insights

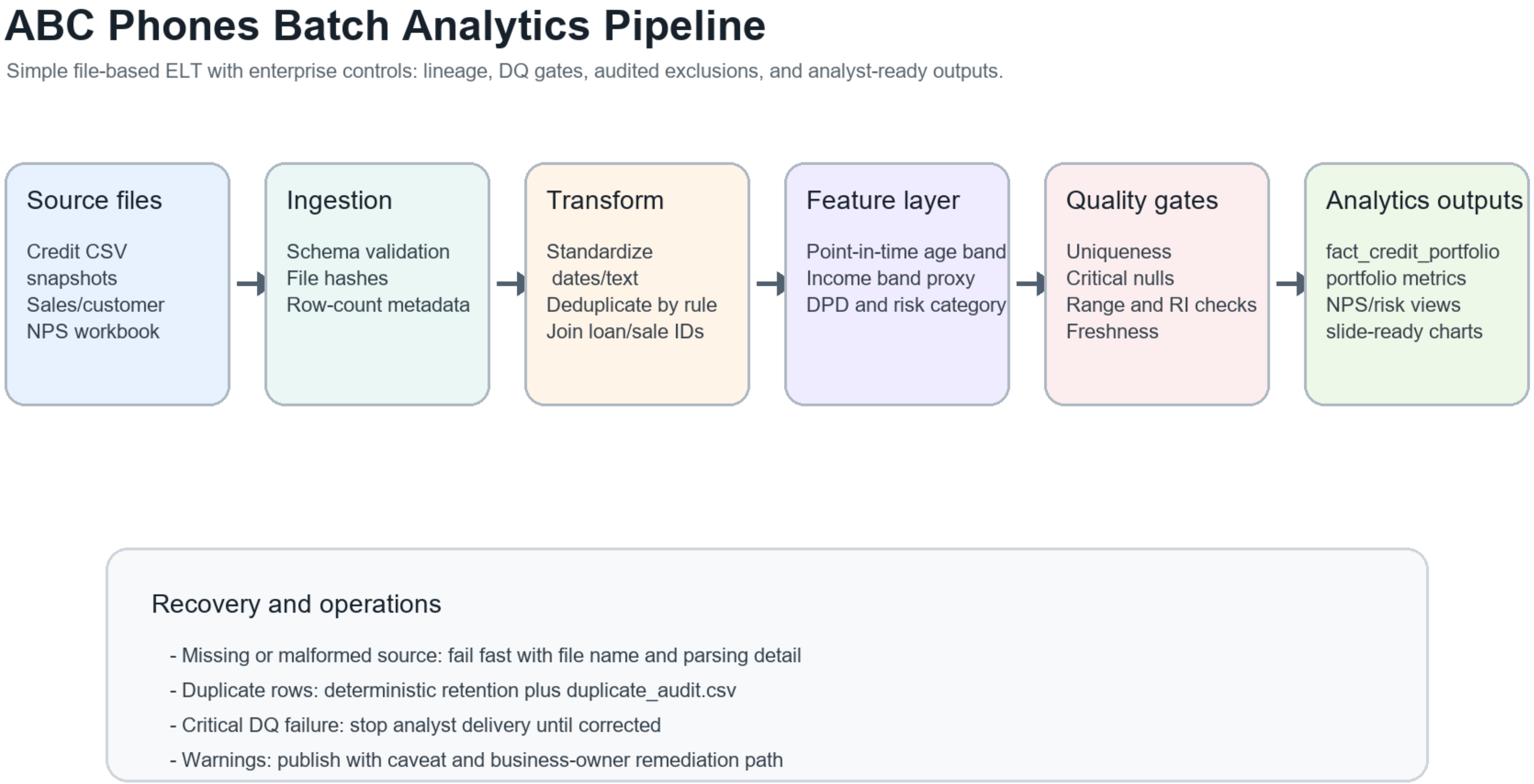
20,740
latest snapshot accounts

38.0%
PAR30 rate

-22.7
critical risk NPS

Pipeline Architecture

A pragmatic file-based design that can graduate to orchestration/dbt/cloud storage without changing business logic.



Engineering Decisions

Small surface area, explicit assumptions, and auditable transformations.

- Credit snapshots drive the grain: one row per LOAN_ID and reporting_date.
- Point-in-time features are recalculated per snapshot rather than copied from the latest view.
- Duplicates use deterministic business rules; discarded rows remain in duplicate_audit.csv.
- Demo accounts are excluded from portfolio analytics and retained in the audit trail.
- Income Level is treated as the monthly-income proxy because employment duration is absent.

20,740
analytics rows

2
DQ warnings

0
critical DQ failures

Data Quality Findings

The pipeline passes critical controls; missing demographic fields are the main business caveat.

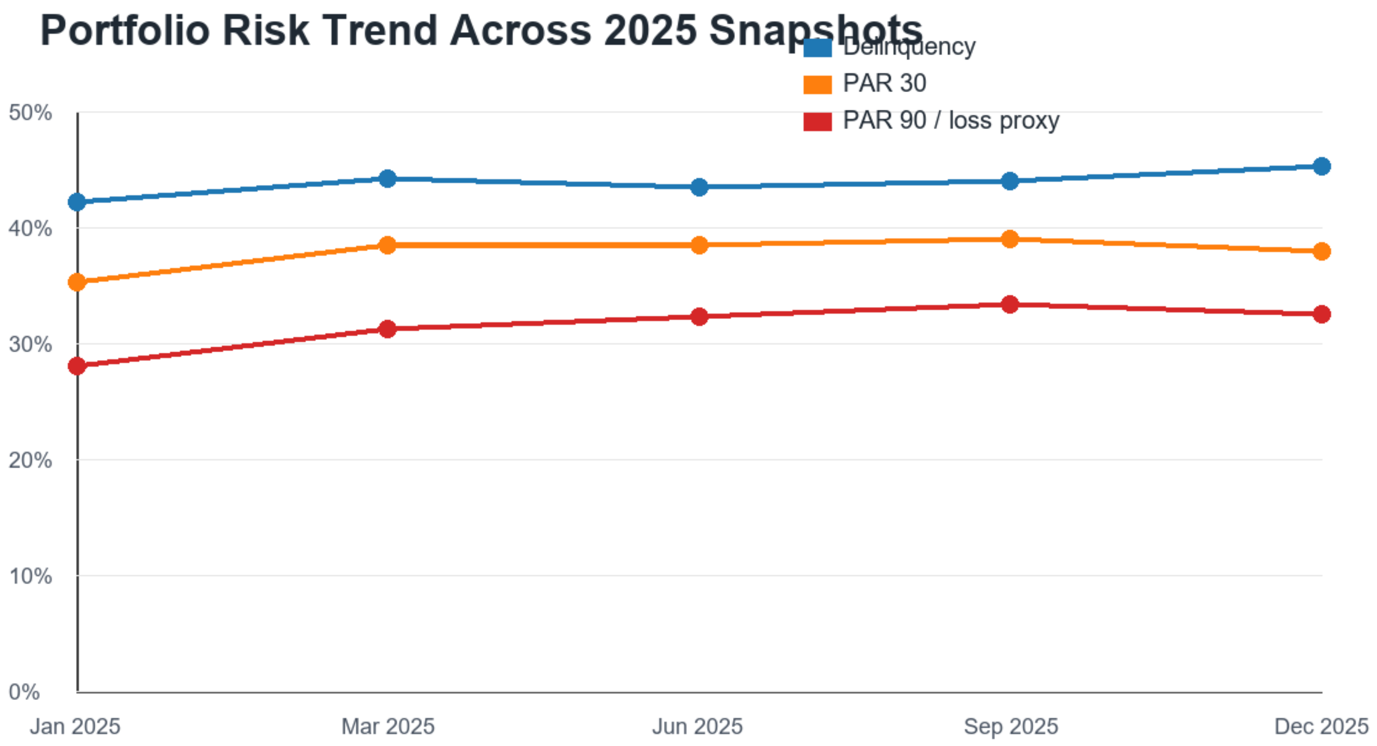
- Uniqueness, critical null, row-count reconciliation, freshness, and referential integrity checks pass.
- DoB missingness is high, pushing many accounts into Unknown age bands.
- Income Level missingness is high, limiting affordability segment
- NPS duplicates are resolved to the latest survey response per l

Controls implemented

- Freshness
- Uniqueness
- Referential integrity
- Range checks
- Null thresholds

Portfolio Health

Delinquency remains elevated throughout 2025; PAR90 is the key loss-proxy watch item.



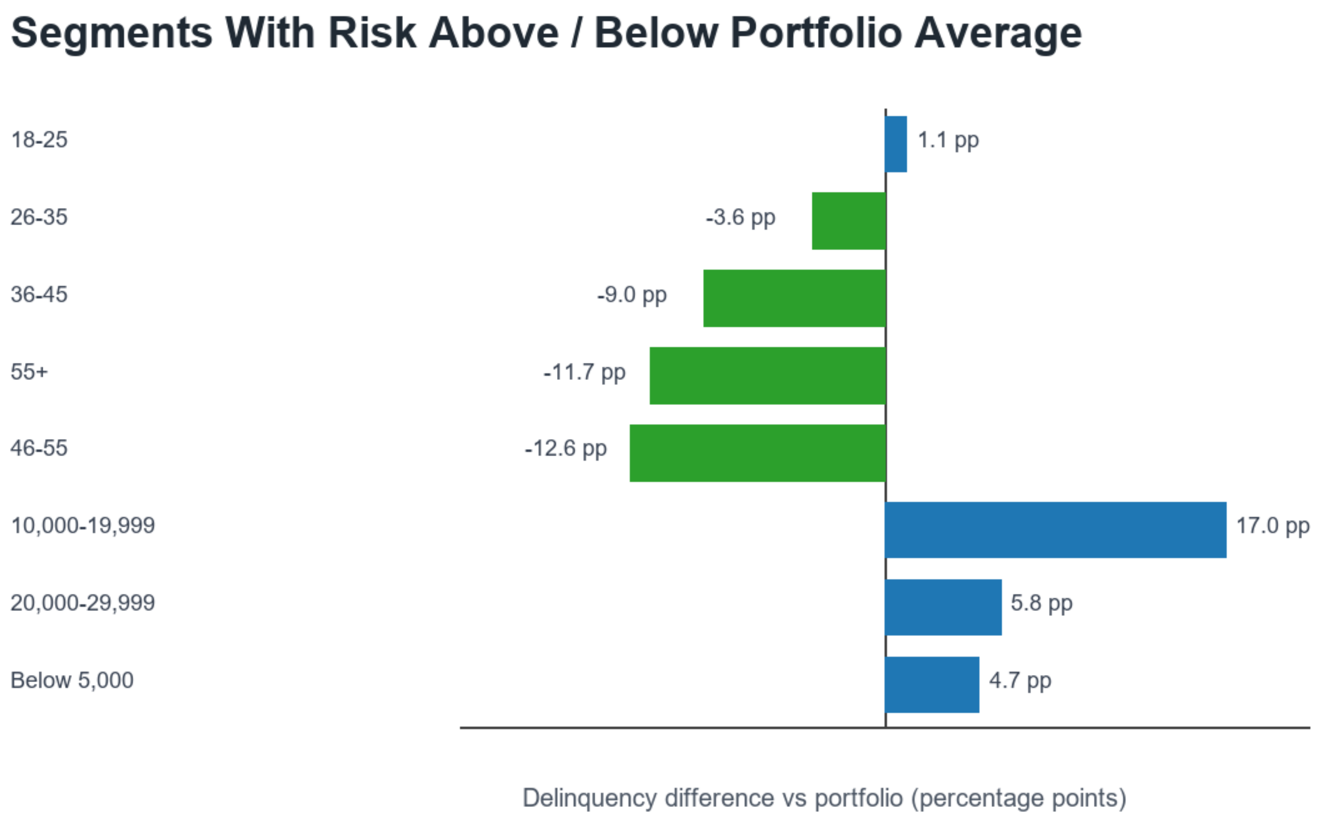
45.3%
latest delinquency

38.0%
latest PAR30

45.5%
arrears / balance

Segment Risk

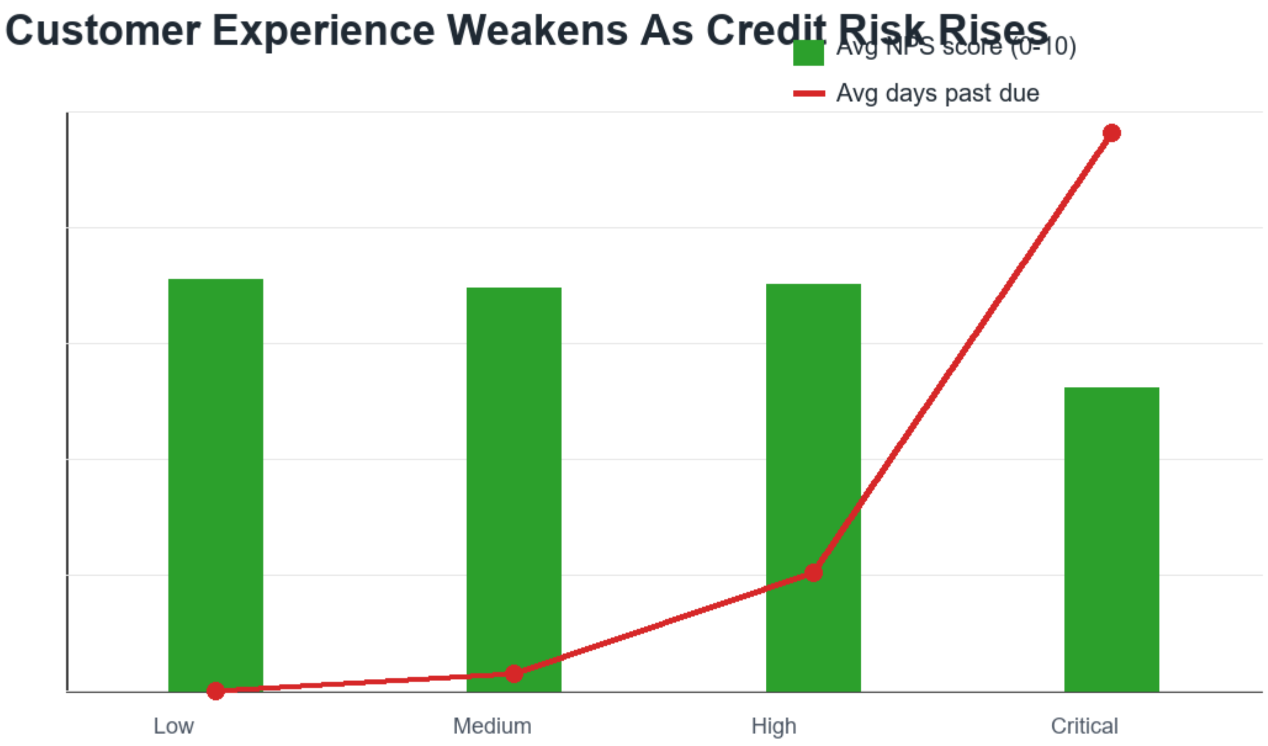
Risk segmentation is useful, but missing demographics must be fixed before using it for policy decisions.



- Most robust elevated segment: 100,000-149,999 (949 acc
- Delinquency runs 3.6 percentage points above portfolio av
- Small low-income bands also show stress, but sample siz

Credit Outcomes x Customer Experience

Critical-risk customers report materially weaker satisfaction.



- Critical-risk average NPS score: 5.2/10.

- Critical-risk NPS index: -22.7.

- Average critical-risk DPD: 260 days.

- This is where collections friction and customer experience

Recommendation

Improve risk and experience together before accounts cross PAR30.

Launch an early-arrears care journey

- Trigger on 1-30 DPD plus rising arrears, before write-off behaviour ha
- Send payment-reflection confirmations and reminders through prefer
- Route payment-lock and support issues into fast resolution before co
- Track the impact using PAR30 migration, payment recovery, repeat c

Data capture upgrades

- Employment tenure and type
- Transaction-level payment ledger
- Standardized customer master
- Collection contact outcomes